POF Workbench Practical Guide

Developed by the Planetary and Artificial Intelligence Laboratory at Southern University of Science and Technology, POF Workbench combines image annotation, model training, active learning and batch inference in one desktop application. Edit labels directly, train a segmentation model from paired data, refine it through reviewed regions, and process new image collections.

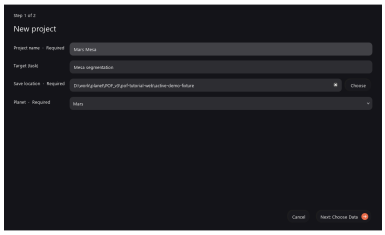
1. Projects and data 2. Annotation and label editing 3. Training 4. Active learning 5. Batch inference 6. Models, DEM and settings 7. Troubleshooting

1. Projects and data

A project groups research work. Each task keeps its own image sources, labels, models and active-learning rounds.

1. Choose **New project** and enter the name, location, target and planet. Use **Switch project** to open an existing project.
2. Choose **Enter task** in the task list. Opening a project does not automatically select a task.
3. Select **Active learning**, **Training** or **Inference**, add the required data and run **Check data**.

Training images and labels must have matching names and pixel dimensions. Unmatched files appear in the check details; valid pairs can still be used. Label normalization creates separate copies and preserves the originals.



Project and task entry. Select a task before preparing data.

2. Annotation and label editing

1. Choose **Start annotation**, then **Open image** or **Open folder**. TIFF and common image formats are supported.
2. Choose an **Output folder**. Existing same-name PNG masks or SHP labels can be loaded for editing.
3. Draw along the target boundary. Press **Ctrl+S** to save, or choose **Save and next**.

Select the output format in Settings: PNG mask is the default; SHP is also available. When both formats exist for one image, the current output format takes priority. SHP requires a valid spatial reference; PNG masks retain the image pixel dimensions.

Editing tools

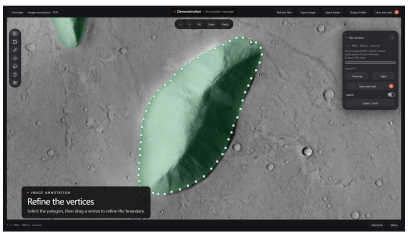
- **Select**: select a polygon, drag its vertices, or delete it.
- **Polygon**: click boundary points and double-click to close. Erase mode removes the enclosed area.
- **Brush / Eraser**: drag to paint or erase. Adjust size with the vertical slider, plus/minus controls or numeric input.
- **Lasso**: draw a freehand outline to add or erase a region.
- **Labels**: toggle visibility without deleting labels. Use **Undo / Redo** to reverse edits.

The file monitor shows position, saved state and file progress. Hover over a tool to see its shortcut; review or change bindings in the shortcuts window.

Frequently used shortcuts

- Tools: **V** select, **P** target polygon, **B** polygon erase, **R** brush, **E** eraser, **L** lasso, **Shift+L** lasso erase.
- Edit and save: **Enter** close polygon, **Esc** cancel drawing, **Delete** delete selection, **Ctrl+Z** undo, **Ctrl+Shift+Z** or **Ctrl+Y** redo, **Ctrl+S** save.
- View: **Space + drag** pan, wheel zoom, **F** fit, **1** actual size, **H** toggle all labels. **- / =** decrease or increase brush size.
- Navigation: **A / D** in standalone annotation; **[/]** in active-learning review. Save with Ctrl+S or Save and next when switching.

Press **F1** to customize bindings and save them for both annotation pages. Hover over a button to see its current shortcut. While a text or numeric input has focus, keys edit the field; click the canvas before using tool shortcuts.



Dark English annotation interface.

3. Training

1. Select image and label folders in the task data page. Run **Check data** and enter **Training** after verifying valid pairs.
2. Select an initialization model and training steps. Adjust Batch, input size and device in advanced settings. Use built-in **POFBase** when no task model is available.
3. Use **Add DEM** if available, or leave it empty. Confirm the device and settings, then choose **Start training**.
4. Watch Loss, validation Dice/F1 and fixed examples. Review the best model and result details before using it for prediction or active learning.

Loss measures training error. Validation Dice/F1, IoU, Precision and Recall describe segmentation quality; inspect example boundaries as well as numbers. The best model is selected by validation performance and may come from an earlier step.

Use **Safe stop** and wait for completion when interrupting a run. If a model was saved but post-training testing did not finish, use the follow-up test action rather than retraining.

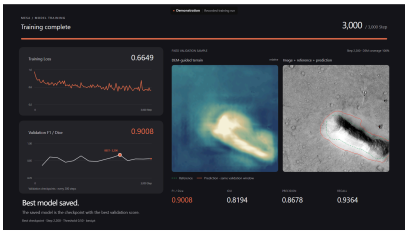
Choosing training parameters

- **Steps**: optimizer updates, not passes through the entire dataset. The regular training page starts at 3,000. Keep this initially and use validation results to decide whether further training helps.
- **Batch**: samples processed together per update; initially 8. If GPU memory runs out, try 4, 2, then 1. A smaller batch is a practical memory adjustment, not proof of worse quality.
- **Image size**: network input side length, initially 384 px. It differs from the field of view sampled from the original image, which may be resized for the network. Increasing input size increases memory and computation.
- **Device**: use auto normally, cuda with a configured compatible NVIDIA environment, or cpu for data checks. CPU training may be much slower.
- **Initialization / Weights**: use POFBase to establish a task model or an existing complete model for further learning. Confirm the weight architecture before starting.

Learning rate and validation intervals come from the training configuration and normally do not need adjustment for the first run. Frequent validation adds runtime. When editing a configuration, keep a copy and change only a few settings at once.

Interpreting progress

A falling Loss means the training error is decreasing; it does not by itself establish better generalization. Check validation Dice/F1, IoU and fixed example boundaries. Low Precision often accompanies false detections; low Recall often accompanies missed targets. If validation deteriorates while training Loss falls, inspect overfitting and the data split before adding more steps.



Training status, validation curves and fixed examples.

4. Active learning

Active learning focuses manual review on recommended regions, then updates the model with confirmed labels. It requires an accessible unlabeled image pool.

1. Add an unlabeled image folder to the task, then enter **Active learning**. Choose an existing starting model or use the task model selected automatically.
2. With a trained model, recommend regions and correct its suggestions. With POFBase, complete initial manual annotations before training the first task model.
3. Enter the review canvas. The orange dashed boundary marks the review area. Adjust it and check every target inside. Use **Terrain** when terrain assistance is available.
4. Correct the suggested labels and choose **Submit and next**. You do not need to finish every region: **Train with completed annotations** uses the completed subset.
5. After training, review the round model, metrics and examples in the overview, then start another recommendation round.

On submission, unmarked pixels inside the reviewed area become background; pixels outside remain unknown. Unreviewed regions are not automatically treated as background. Compare rounds under the same validation conditions.

Round settings and recommendation budgets

- **Initial steps / Round steps**: training length for the initial model and later updates; both initially 1,000. With an existing model, review a small set first and use validation changes to guide further training.
- **Regions per round**: desired review regions, initially 30. Reduce it to match your available review time. You do not need to finish all recommended regions before training.
- **Candidate limit / Max source images / Max model inputs**: bound candidate count, visited images and network inputs. These control recommendation work, not the final label count. Keep large collections within a budget and inspect status records before increasing limits.
- **Recommendation time budget**: allowed recommendation time, initially 180 seconds. A small budget can yield fewer candidates; a larger budget increases waiting time.
- **Window size**: metres for projected imagery or pixels for ordinary imagery. Include the entire target and enough context; do not reduce a large landform to a small boundary fragment.
- **Context margin**: surrounding context, initially 25%. More context can help boundary review, but does not mark pixels outside the review area as confirmed.

MarsScope is optional semantic assistance for finding candidate landforms. Without it, use the available base preparation and review flow. Always check suggestions for both missed targets and incorrect background.



Active learning review canvas. Tutorial gains are illustrative, not performance claims.

5. Batch inference

1. Open **Inference**. In **Input / output**, confirm the complete segmentation model and input/output folders.
2. Add DEM if needed. Choose one output folder per image or grouped folders for mask, Shapefile and other formats.
3. Scan the images, check the queue and choose **Start prediction**.
4. Inspect completed boundaries at a larger scale. To correct results, open the source image and same-name label in the annotation page.

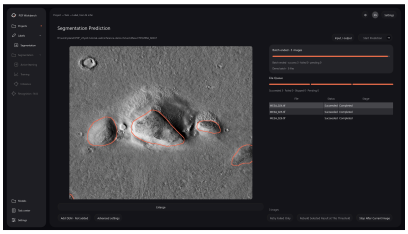
Inference uses the scale windows stored with the model. Target-size filtering removes results without changing model training. PNG masks support further editing; spatial vector outputs can be checked in GIS.

Read failures beside the file or in task records. After checking paths, storage and model availability, use **Retry failed only**. Supported saved results can be regenerated after changing the threshold.

Threshold, windows and output choices

- **Threshold**: converts foreground probabilities to regions. Selecting a model normally loads its saved threshold. Lower values may recover missed targets while adding false positives; higher values are stricter and may miss weak targets. Start with the model value and check representative images before a batch.
- **Target size**: filter regions by area or size. Leave a bound blank for no limit in that direction. Confirm valid spatial referencing before using physical units.
- **Model windows**: normally keep the saved scale windows and weights. Target-size filtering is separate; filtering cannot recover context the model never observed.
- **Continuous / cross-image inference**: for suitable neighbouring images with compatible spatial references. Being in one folder is not sufficient; start with ordinary batch inference for independent images.
- **Output layout**: per-image folders organize individual results; grouping by format simplifies opening masks or Shapefiles together. Confirm the output directory and free disk space first.

Progress distinguishes the current file stage from the overall queue. Multiscale prediction, fusion and export can all take time; remaining time is re-estimated from actual speed. Stop after current image still lets the current file finish.



File status and processing stages in the inference queue.

6. Models, DEM and settings

- **Models**: import a complete model, check its source and validation conditions, then use it for annotation, training or inference. Defaults prioritize the selected best model for the task; a starting model can also be chosen manually.
- **DEM**: optional. If empty or unavailable for the current window, supported POF models fall back to monocular relative terrain. Relative terrain is not measured elevation.
- **Settings**: change language and appearance, including system theme. Download GPU runtime and MarsScope only when needed, then check the configured status.
- **Task center**: inspect stages, errors and full records. Leaving a page does not stop a task.

7. Troubleshooting

- **Active learning cannot start:** enter a task and configure an unlabeled image pool. Use Training when you only have paired training data.
- **Labels do not load:** check names, pixel dimensions and the current output-format priority.
- **GPU unavailable or out of memory:** check runtime configuration and reduce Batch. Use CPU to verify the data workflow if needed.
- **Missing file or output failure:** check network access and free disk space. Move deeply nested Windows projects to a shorter local path.
- **Unexpected metrics:** inspect label foreground and fixed validation examples for empty labels, wrong pairs or class-encoding errors.

Report reproduction steps, task records and the complete error message. Back up source images, labels and models. Only clear caches that are confirmed to be rebuildable.